INCH-POUND

MIL-L-31007
03 October 1994

MILITARY SPECIFICATION

LINER, AIRCREW, BATTLEDRESS, UNIFORM (COAT AND TROUSERS)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for a two-piece insulating liner of quilted aramid batting material intended to be worn under the Aircrew Battle Dress Uniform and as a component of the Aircrew Cold Weather Clothing System.

1.2 Classification. The liner shall be in the following sizes and lengths, as specified (see 6.2).

Schedule of sizes

Short	Regular	Long
Extra-Small	Extra-Small	Extra-Small
Small	Small	Small
Medium	Medium	Medium
Large	Large	Large
Extra-Large	Extra-Large	Extra-Large

NOTE: The sizes (Extra-Small, Small, Medium, Large, and Extra-Large) may be abbreviated as XS, SM, MED, LG, and XLG. The lengths (Short, Regular, and Long) may be abbreviated as S, R, and L.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, ATTN: DPSC-FSSD, 2800 South 20th Street, Philadelphia, PA 19101-8419, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 8415

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

DDD-L-20	-	Label: For Clothing, Equipage, and Tentage (General Use)
JJ-W-155	-	Webbing, Textile (Cotton, Elastic)
V-B-871	-	Button, Sewing Hole; and Button, Staple (Plastic)
A-A-50195	-	Thread, Aramid
A-A-50198	-	Thread, Gimp, Cotton, Buttonhole
A-A-50199	-	Thread, Polyester Core, Cotton- or Polyester-Covered
A-A-55126	-	Fastener Tapes, Hook and Loop, Synthetic
A-A-55195	-	Thread, Para-Aramid, Spun, Intermediate Modulus

MILITARY

MIL-C-508	-	Cloth, Oxford, Nylon, 3 Ounce
MIL-C-3735	-	Cuffs, Knit, Wrist & Ankle, and Cloth, Knitted
MIL-F-10884	-	Fasteners, Snap
MIL-B-41826	-	Batting, Synthetic Fibers: Polyester (Unquilted and Quilted)
MIL-T-43566	-	Tape, Textile, Cotton or Polyester, General Purpose, Natural or in Colors
MIL-T-43685	-	Webbing, Tape, Textile, Aramid Fiber
MIL-C-43774	-	Cloth, Plain or Pajama Check Weave, Aramid
MIL-B-81813	-	Batting, Aramid, or Novoloid Fiber, Quilted

STANDARDS

FEDERAL

FED-STD-595	-	Colors
FED-STD-751	-	Stitches, Seams, and Stitchings

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MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation. (Any additional information is available from the contracting activity or as directed by the contracting officer.)

Department of the Army Technical Manuals

FM-10-280 - Field Laundering

(Copies of documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

RULES AND REGULATIONS UNDER THE TEXTILE FIBER PRODUCTS IDENTIFICATION ACT

(Copies may be obtained without charge from the Federal Trade Commission, Washington, DC 20580-0001.)

U.S. POSTAL MANUAL

(Copies of the manual may be obtained from the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402-0001.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD

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adopted shall be those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM-D 1974 - Practice for Methods of Closing, Sealing and Reinforcing Fiberboard Shipping Containers

ASTM-D 3951 - Standard Practice for Commercial Packaging

ASTM-D 5118 - Practice for Fabrication of Fiberboard Shipping Boxes

(Applications for copies should be addressed to:)
American Society For Testing and Materials
1916 Race Street, Philadelphia, PA 19103-1187

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

THE COLOR ASSOCIATION OF THE UNITED STATES

Department of Defense Standard Color Card for Sewing Threads

Department of Defense Standard Shades for Buttons

Standard Shades for Heraldic Yarns for Embroidery and Ribbons

(Applications for copies should be addressed to:)
The Color Association of the United States
409 W. 44th St., New York, NY 10016-0927

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document; however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3), in accordance with 4.3.

3.2 Samples. Samples, when furnished, are solely for guidance and information to the contractor (see 6.4). Variation from this specification may appear in the sample, in which case this specification shall govern.

3.3 Material. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3.1 Basic material. The basic material shall be batting, aramid or Novoloid fiber, quilted, conforming to MIL-B-81813. The color of the outercovering cloth shall be Olive Green 106 or (Camouflage Green 483).

3.3.2 Binding, curtain, and waistband lining. The material shall be 3 ounces per square yard aramid cloth conforming to MIL-C-43774. The color shall be Olive Green. The finished width shall be 3/8 inch.

3.3.3 Knitted wrist and leg cuffs. The wrist and leg cuffs shall be a one by one rib knit aramid. The wrist cuffs shall conform to type IV, class 2, size 2 of MIL-C-3735. The knit leg cuffs shall conform to type IV, class 3 of MIL-C-3735. The color shall be Olive Green 106. The cuffs shall be doubled to meet the finished length measurements. The wrist cuffs shall finish 4 to 4-1/4 inches in length, and the opening shall finish 2-3/4 to 2-7/8 inches. The leg cuffs shall measure 2-3/4 to 3 inches in length and the opening shall measure 6 to 6-1/4 inches.

3.3.4 Thread. The thread shall be aramid, Camouflage Green 483, matching color chip 34094 of FED-STD-595, approximate tex of 40 or 60 conforming to A-A-50195 or para-aramid, nominal tex 39, 3 ply thread conforming to A-A-55195 .

3.3.5 Fastener tape. The hook and loop fastener tape shall be Green 3421, one inch wide conforming to type I, class 2 of A-A-55126. Hook and loop shall show no sign of fraying.

3.3.6 Labels. Each coat and trouser liner shall be individually bar-coded and have a class 2 size label, a class 1 identification label, and a class 3 instruction label conforming to type VI of DDD-L-20 -- Label: For Clothing, Equipage, And Tentage (General Use). The size and identification labels may be combined into one label, or the size, identification and instruction labels may be combined into one label, type IV, class 14 conforming to DDD-L-20. The

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color shall be medium Green Cable No. 70034, 70130, or 70131 and the colorfastness to laundering shall apply.

3.3.6.1 Size label. The size label shall contain the Size and Stock No. of each coat and trouser liner. Descriptive information shall be printed in letters and numerals not less than 10-point.

3.3.6.2 Instruction label. The instruction label shall contain the following information:

LINER, AIRCREW BATTLEDRESS UNIFORM:

1. Wear under the Aircrew Battledress Uniform for extra warmth.

2. Dry Liner and Aircrew Battledress Uniform separately when damp or wet.

3. Wash by hand or machine launder on permanent press cycle in warm water with a mild laundry detergent. Rinse thoroughly and drip or machine dry at medium heat. Use a fabric softener in either the wash or the dry cycle to reduce static electricity.

4. Field launder in accordance with Natick Formula IX of FM-10-280 or for post laundry, Natick Formula X.

5. DO NOT USE STARCH OR BLEACH.

3.3.6.3 Label placement. The coat liner size label shall be stitched on four sides to the inside center of the back, starting 1 to 1-1/2 inches below the neck edge. The trouser liner size label shall be stitched to the waistband casing joining seam on the center back 1 to 1-1/2 inches below finished waist. The identification label shall abut the bottom edge of the size label or be caught in the stitching. The top edge of the instruction label shall abut the bottom edge of the identification label or be caught in the stitching.

3.3.6.4 Combination label. The size and identification labels may be combined into one label, or the size, identification and instruction labels may be combined into one label, type IV, class 14 conforming to DDD-L-20. If a combination size and identification label is used, it shall be positioned 1 to 1-1/2 inches below the waist and neck edge, with the instruction label positioned abutted to the bottom edge of the identification label or caught in the stitching. If a combination size, identification, and instruction label is used it shall be positioned 1 to -1/2 inches below the finished waist or neck edge.

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3.3.7 Snap fasteners. The snap fasteners shall conform to style 2A, brass, finish 2, male and female complete, consisting of stud and eyelet and socket eyelet (uncapped button) sizes 1 or 2 conforming to MIL-F-10884.

3.4 Design.

3.4.1 Coat liner. The quilted hip length coat liner shall be collarless, with long sleeves, knit cuffs, and a bottom curtain that is 7-1/4 to 7-1/2 inches wide. The material for the curtain extension shall conform to MIL-C-43774. The coat front opening closure shall have 5 hook and loop fastener pieces measuring two by one inches placed horizontally, starting 3/4 inches from the top edge. The coat liner shall close left over right side with the loop tape on the left side and the hook on the right side, aligned with the front edge of the binding and spaced 3-1/2 to 3-5/8 inches apart on the quilted portion of the liner. When fastened, the hook and loop fastener tapes shall be aligned and lay flat and smooth. The sleeves shall be set according to marks on the pattern leaving underarm open area for use as a vent, the edges of this opening shall be bound. The neck and front quilted portions of liner shall be bound. The sleeve setting and curtain seam allowances shall be turned toward the body. All other seam allowances shall be turned toward the coat liner front. (See Figure 1.)

3.4.2 Trouser liner. The trouser liner shall have an elasticized waist with a front waist snap fastener, an open fly with bound edges, and knit leg cuffs. The sewn-on waistband and waistband facing shall be top-stitched 1/8 to 3/16 inch from the top edge. The bottom edge of the waistband casing shall be turned under and top-stitched to trouser liner 1/16 to 1/8 inch from the edge. The elastic shall start and terminate 1-1/2 inches from the trouser front edges and shall be stitched through all plies. The crotch and back joining seam shall be turned in either direction and top-stitched 3/16 to 1/4 inch from the edge. The leg inseam shall be turned toward the front.

3.5 Seams and stitching. Stitching and seams used in the fabrication of the liners are specified in Tables I & II conforming to FED-STD-751. There shall be 10-14 stitches per inch. All LSbm-3 (b) seams shall be stitched 3/16 to 1/4 inch from the edge. No seam or component shall be twisted, puckered, or pleated, and no part of the liners shall be caught in an unrelated seam. Gage of stitching for hook & loop tape shall be 1/8 to 3/16 inch from the edges.

3.6 Patterns. Standard patterns, which provide a seam allowance of 1/2 inch for all joining seams (to finish at least 3/8 inch wide) and 3/8 inch for front extensions, will be furnished by the Government. The Government patterns shall not be altered in any way and are to be used as a guide for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns. Minor modifications are permitted where necessary when using automatic equipment.

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These modifications shall not alter the dimensional, serviceability, or appearance requirements cited in the specification.

TABLE I. Coat Seam and Stitching Chart

	Seam & Stitching Type	Stitch Type
Side, shoulder, and sleeve setting seams	LSbm-3 (a)	515 or 516 or 519
Attaching quilted material to curtain	and LSbm-3 (b)	and 301 or 401
Cuff joining seam	LSbm-3 (a)	515 or 516
Binding	BSc-1	301
Curtain hem	EFd-1	502 or 503
Hook & loop tapes		301

TABLE II. Trouser Seam and Stitching Chart

	Seam & Stitching Type	Stitch Type
Crotch and back seams, inseams, sleeve setting seams	LSbm-3 (a) and LSbm-3 (b)	515 or 516 or 519 301 or 401
Cuff joining seam	LSbm-3 (a)	515 or 516
Top edge of waistband and facing	SSe-2 (a) SSe-2 (b)	301 or 401
Bottom edge of facing	LSd-1	301 or 401
Elastic attachment seam	SSa-1	301 or 401
Binding for fly	BSc-1	301

Note: All LSbm-3 (b) seams shall be stitched 3/16 to 1/4 inches from the edge.

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3.6.1 Pattern parts. The components of this item shall be cut from the material specified in accordance with the pattern parts indicated.

TABLE III. List of pattern parts

Material	Nomenclature	Cut parts
Quilted aramid cloth	Coat front	2
	Coat back	1
	Sleeve	2
Aramid cloth	Curtain	1
Quilted aramid cloth	Trouser leg	2
Aramid cloth	Waistband casing	1

3.7 Construction. The construction shall conform in all respects to the requirements specified in Table V and herein. Figures are furnished solely for guidance and information. Should variation from the specification appear in Figure 1 or Figure 2, the specification shall govern.

3.7.1 Type 301 stitching. Ends of all stitching shall be backstitched or overstitched not less than 1/2 inch except where ends are turned under or caught in other seams or stitching. Thread tensions shall be maintained so that there will be no loose stitching resulting in loose bobbin or top thread, or excesively tight stitching resulting in puckering of the materials sewn. The lock shall be embedded in the materials sewn.

3.7.1.1 Repairs of type 301 stitching. Repairs of type 301 stitching shall be as follows:

a. When thread breaks or bobbin run-out occurs during stitching, the stitching shall be repaired by restarting the stitching a minimum of 1/2 inch back of the end of the stitching. 1/

b. Thread breaks or two or more consecutively skipped or run-off stitches noted during inspection of the item shall be repaired by overstitching. The stitching shall start a minimum of 1/2 inch in back of the defective area on to the existing stitching. Loose or excessively tight stitching shall be repaired by removing the defective stitching without damaging the materials and restitching in the required manner. When damaged thread has been removed needle holes do not constitute damage tc the material. 1/

1/ When making the above repairs the ends of the stitching are not required to be backstitched.

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3.7.1.2 Type 401, 502, 503, 515, 516, and 519 stitching. Thread tension shall be maintained so that there will be no loose stitching, or excessively tight stitching resulting in puckering of the materials sewn. All repairs shall be in accordance with 3.7.1.1a and b. Repairs to stitch type 401 may be accomplished by use of stitch type 301.

3.7.1.3 Repairs. Repairs such as mends, darns, patches, or splices are not permitted on the liner.

3.7.1.4 Automatic stitching. Automatic stitching machines may be used to perform any of the required stitch patterns provided the requirements for the stitch pattern, stitches per inch, and size and type of thread are met, and at least three or more tying, overlapping, or back stitches are used to secure the ends of stitching.

3.7.1.5 Thread ends. All thread ends shall be trimmed to 1/4 inch maximum length.

3.8 Manufacturing operation requirements. The liners shall be manufactured in accordance with the operation requirements specified in Tables IV and V. The contractor is not required to follow the exact sequence of operations listed, unless otherwise specified. Any additional basting or holding stitches used to facilitate manufacture are permissible, provided the thread is removed or does not show on the finished liner.

3.9 Sizes and Measurements. The finished coat and trouser liner shall conform to the measurements in Table IV and V.

Table IV. Coat liner finished measurements (in inches)

	Short	Regular	Long	Tolerance
1/2 Chest 1/				
X-Small	21 1/4	21 1/4	21 1/4	± 3/4
Small	23 1/4	23 1/4	23 1/4	± 3/4
Medium	25 1/4	25 1/4	25 1/4	± 3/4
Large	27 1/4	27 1/4	27 1/4	± 3/4
X-Large	29 1/4	29 1/4	29 1/4	± 3/4
Back Length 2/				
X-Small	31	31 1/2	32	± 1/2
Small	31 1/2	32	32 1/2	± 1/2
Medium	32	32 1/2	33	± 1/2
Large	32 1/2	33	33 1/2	± 1/2
X-Large	33	33 1/2	34	± 1/2
Sleeve length 3/				
X-Small	16 1/2	17 1/2	18 1/2	± 1/2
Small	16 1/2	17 1/2	18 1/2	± 1/2
Medium	16 1/2	17 1/2	18 1/2	± 1/2
Large	16 1/2	17 1/2	18 1/2	± 1/2
X-Large	16 1/2	17 1/2	18 1/2	± 1/2

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1/ Measurement shall be taken with the liner flat and the front closed, folded edge to folded edge at the base of the armhole.

2/ Measurement shall be taken along center back from edge of neck to bottom edge of liner tail.

3/ Measurement shall be taken from top of binding at armhole along underarm seam to bottom of knitted cuff. Cuff measures 4 1/2 inches in width.

Table V. Trouser liner finished measurements (in inches)

	Short	Regular	Long	Tolerance
1/2 Waist 1/				
X-Small	13	15	17	+ 3/4, - 1/2
Small	13	15	17	+ 3/4, - 1/2
Medium	13	15	17	+ 3/4, - 1/2
Large	13	15	17	+ 3/4, - 1/2
X-Large	13	15	17	+ 3/4, - 1/2
Inseam 2/				
X-Small	20 1/4	22 1/4	24 1/2	± 3/4
Small	20 1/4	22 1/4	24 1/2	± 3/4
Medium	20 1/4	22 1/4	24 1/2	± 3/4
Large	20 1/4	22 1/4	24 1/2	± 3/4
X-Large	20 1/4	22 1/4	24 1/2	± 3/4

1/ Measurement shall be taken with the waist snapped, and the liner laid flat and smooth, with elastic relaxed. Measure across front from side edge to side edge.

2/ Measurement shall be taken along inside seam of liner from crotch seam to bottom of leg cuff. Cuff measures three inches in width.

3.10 Workmanship. The finished liner shall conform to the quality of product established by this specification and the occurrence of defects shall not exceed the applicable acceptable quality levels.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements; however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point, or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.2 and 4.4.3.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

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4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

4.4.2 End item visual examination. The coat liners and trouser liners shall be examined separately for the defects listed below. The lot size shall be expressed in units of coat liners or trouser liners as applicable. The sample unit shall be one coat liner or trouser liner as applicable. The inspection level shall be II and the acceptable quality level (AQL), expressed in terms of defects per hundred units, shall be 2.5 for major defects and 40 for total (major and minor combined) defects.

TABLE VI. End item visual defects

Examine	Defect	Classification Major	Minor
Material defects	Any ply having a hole, cut, tear, smash, burn or needle chew	101	
	Any ply having a darn or mend		201
	Exposed drill hole (in any ply)		202
	One or more rows of stitching omitted in the quilting	102	
	More than 1-1/2 inches of broken or missing stitches in the quilting	103	
	Thick or thin place	104	
	Piecing or splicing of outer covering (either ply)	105	
Cleanness	Any spot or stain on outside		203
	One or more size markings not removed		204
	Thread ends not trimmed or loose threads not removed throughout liner, except ends of quilting stitches		205

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TABLE IV. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Component and assembly	Any component part or required operation omitted (unless otherwise classified herein)	106	
	Any operation not performed as specified (unless otherwise classified herein)		206
	The liner, coat and trousers, fabricated natural thread of dumbbell pattern stitching facing wearer	107	
Cutting	Any component not cut in accordance with directional lines on patterns	108	
Seams and stitching (except quilting stitching)	Accuracy of seaming:		
	- Seams, twisted, puckered, or pleated		207
	NOTE: Aramid fabric has a tendency to pucker at joining seams and at stitchings. The puckering when not so excessive as to seriously affect appearance shall not be classified as a defect.		
	- Part of garment caught in unrelated operation or stitching	109	
	- Ends of stitching back-stitched less than 1/2 inch		208
	- Stitching overlapping end(s) of thread break less than 1/2 inch		209
	- Various shades of thread used on outside of same liner		210
	- Ends of a continuous line of stitching overlapped less than 1/2 inch		211

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TABLE IV. End item visual defects (cont'd)

Examine	Classification Defect	Major	Minor
	Gage of stitching and seam allowances:		
	- Irregular, i.e., unevenly gaged or corresponding stitchings not uniformly gaged (to be scored only when the condition exists on major portion of seam)		212
	- Width not as specified or not within range of gage specified or varies more than 6 inches when no range is specified		213
	- Edges or raised stitching sewn too close to edge resulting in damage to the cloth	110	
	Open seams (not applicable to quilting stitching):		
	- Any seam open by:		
	- more than 1 inch	111	
	- more than 1/4 inch up to 1 inch inclusive		214
	NOTE: A seam shall be classified as open when one or more stitches joining a seam are broken or when two or more consecutive skipped or run-off stitches occur. On double stitched seams, a seam is classified as open when either one side or both sides of seam are open.		
	- Raw edges (excluding over-edge stitching and end of binding, when not turned under):		
	- more than 1 inch	112	
	- more than 1/4 inch up to 1 inch inclusive		215
	- Any raw edge caused by omission of overedge stitching for a distance of 3 inches or more		216

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TABLE IV. End item visual defects (cont'd)

Examine	Classification Defect	Major	Minor
	NOTE: Raw edges are classified as such when they occur at places where edge is required to be bound, turned under or overedge stitched. Raw edges not securely caught in stitching shall be classified as open seams.		
	Run-off:		
	- Joining seams - when resulting in an open seam use "Open seam" classification	113	
	- Edge or raised stitching (when not resulting in open seam)		217
	Seam and stitch type:		
	- Seam or stitch type not as specified		218
	- Overedge stitched edge on joining seams not finishing on outside of liner		219
	- Looper thread of 401 stitch type on joining seam finishing on inside of seam (LSbm-3 seam type only)		220
	Stitch tension:		
	- Loose tension resulting in a loose seam		221
	- Loose tension on raised or edge stitching		222
	- Tight tension (stitches break when strain is applied to seam or stitching)		223
	NOTE: Puckering is evidence of tight tension. When puckering is evident, the seam shall be tested		

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TABLE IV. End item visual defects (cont'd)

Examine	Classification Defect	Major	Minor
	by exerting pull in length-wise direction of seam.		
	Stitches per inch (to be scored only when the condition exists on the major portion of seam; except labels):		
	- Three or more, less than minimum, specified	114	
	- One or more stitches, less than minimum specified, on one or more labels		224
	- One or more stitches, more than specified		225
	- Skipped stitches (more than 1/4 inch)		226
Snap fasteners	Any part of assembly missing, mismated, broken, cracked, bent, not securely clinched, affecting function:		
	- two or more snap fasteners	115	
	- one snap fastener		227
	Clinched too tightly cutting surround-ing fabric	116	
	Loose, (i.e., socket or stud spins freely or wobbles in connection portions)		228
	One or more having rough or sharp edge	117	
	More than three splits in eyelet or button barrel		229
Hook & loop fasteners	Hook & loop out of alignment or not able to be fastened properly, width or length of hook & loop not as		

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TABLE IV. End item visual defects (cont'd)

Examine	Classification Defect	Major	Minor
	specified, not sewn 1/8 to 3/16 inch from the edge	118	
Labels (coat and trouser liners)	Size, instruction, or combination label, (when applicable) missing, incorrect, or illegible	119	
	Identification label missing, incorrect or illegible		230
	Any label not stitched on all four sides		231
	Stitching through the printing		232
	Any label off center back by more than 1 inch		233
	Size label (or combination label) less than 1 inch or more than 1-1/2 inches below neck edge or finished waist		234
	Identification label not abutting the bottom edge of the bottom edge of the size label or caught in the stitching		235
	Instruction label not abutting the bottom edge of the identification label or caught in the stitching		236
Binding	Stitching not caught in fabric, (i.e., can be easily pulled away from the edge of fabric)	120	
	End of binding not turned under and caught in the stitching or turned over binding to the outside of liner and tacked		237

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TABLE IV. End item visual defects (cont'd)

Examine	Classification Defect	Major	Minor
	Binding stretched, i.e., edges have a wavy appearance		238
	Finishing less than 1/2 or more than 5/8 inch wide		239

4.4.3 End item dimensional examination. The coat liners and trousers liners shall be examined separately for conformance to measurements specified in Table IV and V. Sleeve or leg lengths unequal by more than 1/2 inch or any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of coat liners or trouser liners. The sample unit shall be one coat liner or trouser liner. The inspection level shall be S-3 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.4 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2, and the AQL, expressed in terms of defects per hundred units, shall be 2.5.

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as: incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container
Content	Size shown on one or more coats or trousers not as specified on outer container

4.4.5 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1, and the AQL, expressed in terms of defects per hundred units, shall be 6.5.

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Examine	Defect
Finished dimension	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Load not bonded as specified
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial (see 6.2).

5.1.1 Level A. Each coat or trouser shall be folded to approximately 11-1/2 by 9-1/2 inches.

5.1.2 Commercial. Coats or trousers shall be preserved in accordance with ASTM-D 3951.

5.2 Packing. Packing shall be level A, B or Commercial (see 6.2).

5.2.1 Level A. Twenty coats or twenty trousers shall be packed in a fiberboard box conforming to style RSC-L, grade V2s of ASTM-D 5118. Inside dimensions of the box shall be approximately 23-3/4 by 19-3/4 by 10 inches. Each box shall be fitted with a plastic case liner, which shall be closed after filling with garments. Each box shall be closed, sealed and reinforced in accordance with ASTM-D 1974.

5.2.2 Level B. Twenty coats or twenty trousers shall be packed in a fiberboard box conforming to style RSC-L, grade V3c of ASTM-D 5118. Each box shall be fitted with a plastic case liner, which shall be closed after filling with garments. Each box shall be closed and sealed in accordance with ASTM-D 1974.

5.2.3 Commercial. Coats or trousers shall be packed in accordance with ASTM-D 3951.

5.3 Palletization. Boxes shall be palletized in accordance with MIL-STD-147 on a 4-way entry pallet. Pallet pattern shall be number 90.

5.4 Marking. Marking shall be in accordance with MIL-STD-129.

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6. NOTES

(This Section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The liner is intended for use by military personnel of the Department of Defense as a insulating liner for the Aircrew Battledress Uniform Trousers and Coat (Cold Weather Clothing System).

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification.
b. Size required (see 1.2).
c. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
d. Whether coat or trouser liner, or both are required.
e. When a first article is required (see 3.1, 4.3, and 6.3).
f. Levels of preservation and packing (see 5.1 and 5.2).

6.3 First article. When a first article sample is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

6.4 Samples. For access to samples, address the contracting activity issuing the invitation for bids or request for proposal.

6.5 Subject term (key words) listing.
Aircrew Battle Dress Uniform
Aircrew Cold Weather Clothing System
Cold weather clothing
Insulation
Liner (coat and trousers)

Custodian:
Army-GL

Preparing Activity:
DLA-CT

Review Activity:
Army-MD

Project Number:
8415-0973

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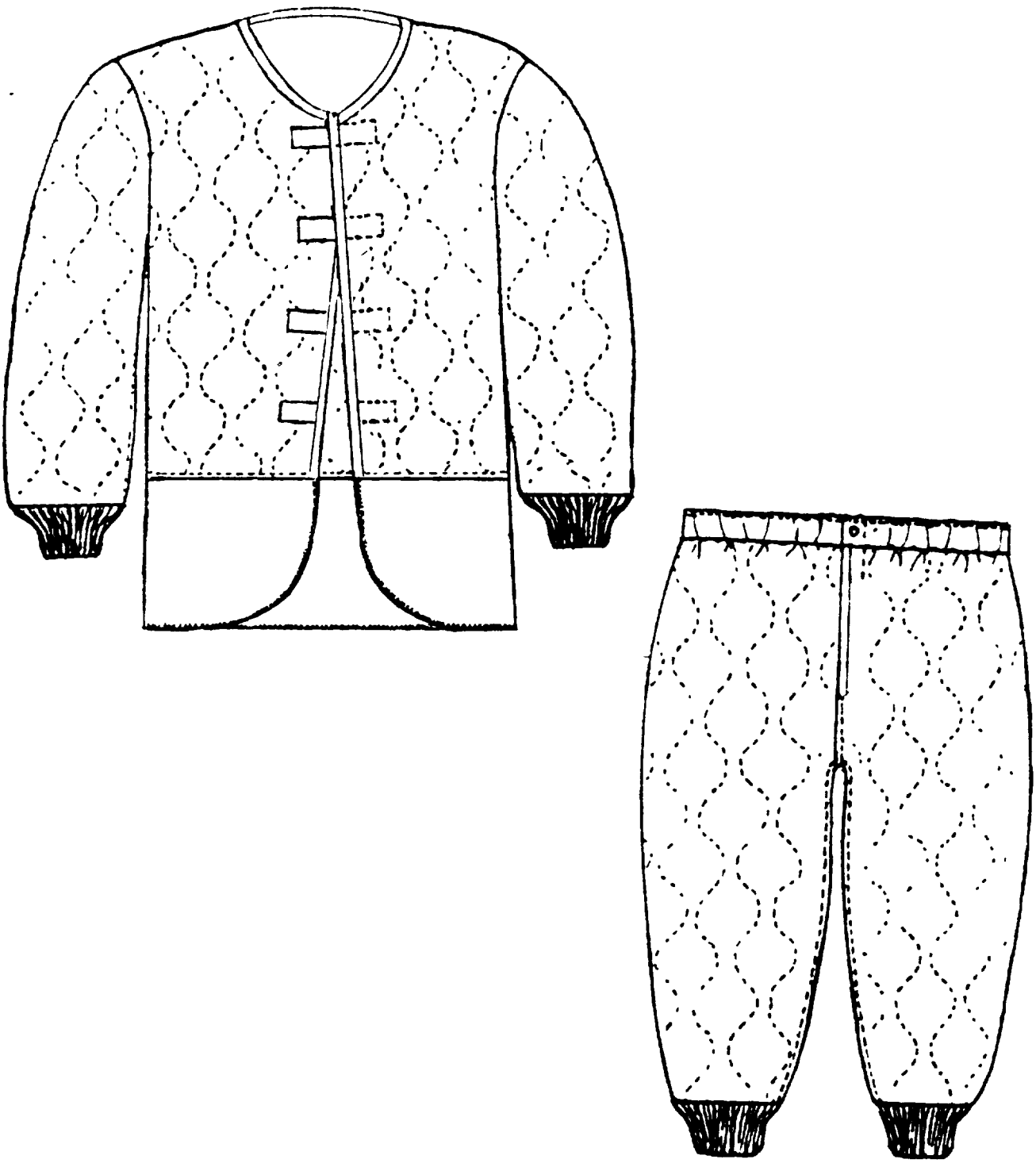


FIGURE 1

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8 In block 1, both the document number and revisi letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification requirements on current contracts Comments submitted on this form do not constitute or imply authorization waive any portion of the referenced document(s) or to amend contractual requirements

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-L-31007	94-10-03

3. DOCUMENT TITLE: LINER, AIRCREW, BATTLEDRESS, UNIFORM (COAT AND TROUSERS)

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5 REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8 PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code)	
Defense Personnel Support Center ATTN: DPSC-FSSD (12-3-D) DLA-CT	(1) Commercial (215) 737-8105	(2) AUTOVON 444-8105
c. ADDRESS (Include Zip Code) 2800 South 20th Street P. O. Box 8419 Philadelphia, Pennsylvania 19101-8419	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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